

SaaS Subscription Analytics

Power BI Dashboard — Project Completion Summary

Tool	Power BI v2.153.910.0 (April 2026)
Pages Built	6 Report Pages
DAX Measures	20+ Custom Measures
RLS Roles	4 Security Roles
Completed	May 2026

Dashboard Pages Overview

#	Page Name	Key Visuals	Status
1	KPI Cards (Executive)	ARR, MRR, NRR, Churn cards + Map + Line chart	COMPLETE
2	MRR Waterfall Bridge	7-bar waterfall + Data labels	COMPLETE
3	Churn Analysis	Trend line + Bar chart + Donut + Slicers	COMPLETE
4	Segmentation	Matrix + Scatter + Stacked bar	COMPLETE
5	Plan Mix	Stacked col + 100% stacked + Treemap + Cards	COMPLETE
6	RLS Demo	Executive view + Region table + Role cards	COMPLETE

Page 1 — KPI Cards (Executive)

The executive overview page providing top-level SaaS metrics at a glance with geographic distribution and MRR trends.

Visuals Built

- 4 KPI Cards: ARR (23M), MRR (2M), NRR % (1.07), Churn Rate % (54.55%)
- World Map: Customer distribution by country
- Line Chart: MRR and MRR 3M Avg by plan (Enterprise/Growth/Pro/Starter)
- Year Slicer: 2021, 2022, 2023, 2024

Key Decisions

- Used fact_mrr_monthly[mrr] as base MRR measure via SUM()
- dim_date connected as time intelligence table
- Map visual used dim_customers[country] for geographic distribution
- MRR 3M Avg calculated using DATESINPERIOD() for rolling average

Page 2 — MRR Waterfall Bridge

The most complex page — a 7-bar MRR waterfall showing the complete revenue bridge from Starting MRR to Ending MRR with all movement types.

Visuals Built

- 7-bar Waterfall: Starting MRR → New MRR → Expansion → Contraction → Churned MRR → Ending MRR → Total
- Data labels on every bar showing exact values
- Color coding: Green (increases), Red (decreases), Blue (totals)
- Month slicer for time filtering

Key Challenges Solved

Challenge	Solution
auto_renew column was Boolean not Integer	Used FALSE() instead of = 0 in DAX filter
fact_mrr_monthly had no movement type column	Used EXCEPT() pattern comparing month-over-month customer lists
Circular dependency between [New MRR] and [Churned MRR]	Churned MRR derived mathematically: [MRR LM] + [Expansion] - [Contraction] - [MRR]
Bridge table exploded when added to Breakdown field	Created via Enter Data (static table) instead of DAX calculated table
Month sort order wrong (descending)	Sorted Category column by SortOrder numeric column in Data view

Core DAX Measures

```
WF_MRR = SWITCH(SELECTEDVALUE('Waterfall MRR'[Category]),
  "Starting MRR", [MRR LM],
  "New MRR",      [New MRR],
  "Expansion",    [Expansion MRR],
  "Contraction",  -ABS([Contraction MRR]),
  "Churned MRR",  -ABS([Churned MRR]),
  "Ending MRR",   [MRR], BLANK())
```

```
New MRR = -- EXCEPT() pattern: customers in current month NOT in previous month
Churned MRR = [MRR LM] + [Expansion MRR] - ABS([Contraction MRR]) - [MRR]
```

Page 3 — Churn Analysis

Deep-dive into customer churn patterns with trend analysis, plan-level breakdown, and reason code distribution.

Visuals Built

- 4 KPI Cards: Churned Customers (600), Churn Rate % (54.55%), Total Refund (8.65K), Reactivation Rate % (10.17%)
- Line Chart: Churn trend Jan=21 growing to Dec=133
- Bar Chart: Churn by Plan Tier (Pro=225, Enterprise=195, Basic=180)
- Donut Chart: Churn Reason Breakdown (features 19%, budget 17.33%, support 17.33%, unknown 15.83%, competitor 15.33%, pricing 15.17%)
- Slicers: Reason dropdown + Month dropdown

Key Challenges Solved

- USERRELATIONSHIP failed — churn_date had no relationship to dim_date — removed USERRELATIONSHIP, used direct date column on X-axis
- Churn Rate showing 120% — fixed using $\text{DIVIDE}([\text{Churned}], [\text{Total}] + [\text{Churned}])$ — standard SaaS formula
- Line chart was flat (same value every month) — fixed by using ravenstack_churn_events[churn_date] directly on X-axis

DAX Measures

```
Churn Rate % = DIVIDE([Churned Customers], [Total Customers] + [Churned Customers])
Reactivation Rate % = DIVIDE(CALCULATE(COUNTROWS(...), is_reactivation=TRUE()), [Churned Customers])
Total Refund = SUM(ravenstack_churn_events[refund_amount_usd])
```

Page 4 — Segmentation

Customer segmentation analysis showing MRR distribution across industries, LTV vs churn positioning, and geographic spread.

Visuals Built

- Plan Tier Slicer (Basic/Enterprise/Pro tiles)
- Matrix: MRR by Industry x Plan Tier (5 industries x 3 plans with heat map coloring)
- Scatter Plot: LTV vs Churn Rate (5 bubbles sized by Customer Count)
- Stacked Bar: Customer Count by Country and Plan (6 countries: DE, US, AU, GB, CA, IN)

Key Challenges Solved

- dim_customers has no plan_tier column — used ravenstack_accounts[plan_tier] throughout
- Scatter chart showed only 1 bubble — fixed by adding account_id to Values field well
- Matrix showed identical values across all plans — relationship gap between fact_mrr_monthly and ravenstack_accounts (known limitation documented)

Page 5 — Plan Mix

Revenue composition analysis showing how MRR is distributed across plans over time with ARR breakdown.

Visuals Built

- Stacked Column: MRR by Plan Over Time (Jan-Dec, growing from 0M to 3M)
- 100% Stacked Column: Plan Mix Share (Enterprise ~75%, Pro ~20%, Basic ~7%)
- Treemap: ARR by Plan Tier (Enterprise=8M, Pro=2M, Basic=1M)
- 3 KPI Cards: Total ARR (69,146.81K), Total MRR (5,762.23K), Avg MRR per Customer (11.52K)
- Slicers: Month dropdown + Plan tier tiles

Key Challenges Solved

- dim_date[month_name] wouldn't connect to ravenstack_subscriptions — created relationship in Model view: dim_date[date] to ravenstack_subscriptions[start_date]
- X-axis showed 'Count of Month' — fixed by using dim_date[month_name] (text column) instead of date hierarchy
- Month order was descending — fixed by sorting month_name column by numeric month column in Data view

Page 6 — RLS Demo

Row-Level Security demonstration showing how different user roles see filtered data based on their region and access level.

Visuals Built

- Active Role banner (red, prominent at top)
- KPI Cards row (copied from Executive page)
- Data Access by Region table (country, MRR, Customer Count, Churn Rate %)
- Role explanation cards (Admin, RegionManager, Viewer)
- World Map filtered by active role

RLS Roles Created

Role Name	Table	DAX Filter	Access Level
Admin	dim_customers	[customer_id] <> ""	All regions, all data
Region Manager_DE	dim_customers	[country] = "DE"	Germany only
Region Manager_US	dim_customers	[country] = "US"	USA only
Viewer	ravenstack_accounts	[is_trial] = FALSE()	Paid customers only

Complete DAX Measures Reference (20+)

Measure	Formula Pattern	Page Used
MRR	SUM(fact_mrr_monthly[mrr])	All
ARR	[MRR] * 12	All
MRR LM	CALCULATE([MRR], DATEADD(...,-1,MONTH))	Waterfall, KPI
MRR 3M Avg	CALCULATE([MRR], DATESINPERIOD(...))	KPI Cards
MRR YoY %	DIVIDE([MRR]-[MRR LY], [MRR LY])	KPI Cards
ARR	[MRR] * 12	Plan Mix, RLS
NRR %	DIVIDE([MRR], [MRR LM])	KPI Cards
LTV	DIVIDE([Avg MRR per Customer], [Churn Rate %])	Segmentation
Avg MRR per Customer	DIVIDE([MRR], [Total Customers])	Plan Mix
Avg Lifespan Months	DIVIDE(1, [Churn Rate %])	Segmentation
Churn Rate %	DIVIDE([Churned], [Total]+[Churned])	Churn, KPI
Churned Customers	COUNTROWS(ravenstack_churn_events)	Churn
New Customers	COUNTROWS filtered by signup month	Segmentation
Retained Customers	[Total Customers] - [Churned Customers]	Churn
Retention Rate %	DIVIDE([Retained Customers], [Total Customers])	Churn
Total Customers	DISTINCTCOUNT(dim_customers[customer_id])	All
WF_MRR	SWITCH(SELECTEDVALUE(Category), ...)	Waterfall
New MRR	EXCEPT() month-over-month pattern	Waterfall
Churned MRR	[MRR LM]+[Expansion]-[Contraction]-[MRR]	Waterfall
Expansion MRR	CALCULATE([MRR], upgrade_flag=TRUE())	Waterfall
Contraction MRR	CALCULATE([MRR], downgrade_flag=TRUE())	Waterfall
Total Refund	SUM(ravenstack_churn_events[refund_amount_usd])	Churn
Reactivation Rate %	DIVIDE(reactivated count, [Churned Customers])	Churn
Customer Count	DISTINCTCOUNT(dim_customers[customer_id])	Segmentation
Churn Count by Reason	COUNTROWS(ravenstack_churn_events)	Churn

Key Lessons Learned

Waterfall Visual Requires Bridge Table

The Power BI Waterfall chart cannot use a column directly. It requires a bridge/lookup table with categories and a SWITCH measure that returns the correct value per category row.

Boolean vs Integer in DAX

Columns stored as Boolean must use TRUE()/FALSE() in DAX comparisons, not 1/0. Using integers causes a type mismatch error at runtime.

USERELATIONSHIP Limitations

USERELATIONSHIP only works when a physical relationship already exists in the model between the two columns. It cannot create new relationships.

Circular Dependency Prevention

When two measures reference each other, Power BI throws a circular dependency error. Always derive one measure mathematically from the other using the MRR bridge identity.

Month Name Sorting

Text month names (January, February...) sort alphabetically by default. Always sort month_name column by the numeric month column using Column Tools > Sort by Column.

Enter Data vs DAX Calculated Tables

Static lookup tables created via Enter Data work more reliably with the Waterfall visual's Breakdown field than DAX calculated tables using DATATABLE().

Relationship Gaps Cause Silent Failures

Missing relationships between tables cause visuals to return 0 or blank without clear error messages. Always verify relationships in Model view before building visuals.

Cross-Table Aggregations

When MRR is in one table and plan_tier is in another with no relationship, aggregations silently ignore the filter. Create the relationship in Model view first.

Data Model & Schema

Tables & Key Columns

Table	Key Columns	Role
dim_customers	customer_id, company_name, country, industry, is_trial, referral_source	Customer dimension
dim_date	date, month, month_name, quarter, week, year, day_of_week	Date dimension
fact_mrr_monthly	customer_id, month, mrr, country	MRR fact table
fact_subscriptions	customer_id, start_date, end_date, mrr, plan, auto_renew, upgrade_flag, churn_flag	Subscription fact table
ravenstack_accounts	account_id, account_name, churn_flag, country, industry, is_trial, plan_tier, signup_date, referral_source	Account dimension
ravenstack_churn_events	account_id, churn_date, churn_event_id, feedback_text, is_reactivated, churn_reason_code, refund_amount_usd, preceding_churn_events	Churn events

Table	Key Columns	Role
ravenstack_subscriptions	account_id, mrr_amount, billing_frequency, plan_tier, seats, start_date, closed_at, details_renew_flag, upgrade_flag, down	Subscription details
ravenstack_feature_usage	account_id, feature_name, usage_date, usage_count, usage_duration, features_usage_beta_feature, error_count	Feature usage
ravenstack_support_tickets	account_id, closed_at, escalation_flag, first_response_time_minutes, priority, resolution_time_hours, satisfaction_score	Support tickets
Waterfall MRR	Category, SortOrder, Type	Bridge table for waterfall visual
_Measures	All DAX measures (20+)	Measures table

Project Statistics & Time Summary

Metric	Value
Total Pages	6
DAX Measures Created	25+
RLS Roles	4
Tables in Model	11
Relationships Created	8+
Visuals Built	25+
Slicers	10+
Errors Resolved	15+
Power BI Version	v2.153.910.0 (April 2026)
Project Duration	1 day (after 4 days on waterfall alone)

Time per Page (Approximate)

Page	Estimated Time	Complexity
Page 1 — KPI Cards	30 minutes	Low
Page 2 — MRR Waterfall	4+ hours	Very High
Page 3 — Churn Analysis	45 minutes	Medium
Page 4 — Segmentation	45 minutes	Medium
Page 5 — Plan Mix	1 hour	Medium-High
Page 6 — RLS Demo	30 minutes	Low-Medium